

Domain adaptation for Catalan–Basque machine translation via synthetic data and continued fine-tuning

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Abstract

Catalan–Basque (CA–EU) machine translation remains highly under-resourced, particularly in specialized domains. We investigate whether synthetic bilingual data derived from monolingual and pivot-language resources can improve domain-specific CA–EU translation. We also explore whether direct domain-specific supervised fine-tuning or continued supervised fine-tuning from a general-domain checkpoint is more effective. We fine-tune Latxa-Qwen3-VL-8B-Instruct¹ with LoRA adapters on three domains: general, literary, and clinical. Literary and clinical training corpora are constructed via pivot translation and back-translation using Latxa-Llama-3.1-8B-Instruct. Domain-specific fine-tuning yields large improvements over both the zero-shot baseline and the general-domain model in all domains. Direct fine-tuning outperforms continued adaptation from the general checkpoint for both literary and clinical domains. A token-matched comparison (fine-tuning with the same number of tokens in both domains) reveals that clinical texts are more learnable than literary texts, probably because clinical texts are stylistically more similar to the standard-register texts on which LLMs are mostly pretrained.

1 Introduction

Catalan and Basque are two co-official languages of Spain with active institutional support, yet the language pair Catalan–Basque (CA–EU) is among the most under-resourced in the Western European MT landscape. Unlike Catalan–Spanish or Basque–Spanish, CA–EU lacks substantial parallel corpora in any domain, and dedicated MT systems for this pair are scarce. The linguistic distance between

the two languages (Catalan is a Romance language while Basque is an isolated language with a highly agglutinative morphology) makes the transfer from other language pairs non-trivial.

Additionally, state-of-the-art LLMs show promising MT performance in high-resource settings, but they still underperform standard NMT systems in low-resource domains such as medicine (Bawden and Yvon, 2023). This work addresses three core questions. First, can pivot-based corpus construction and back-translation generate effective training data for low-resource specialized CA–EU MT, in the absence of direct in-domain parallel corpora? Second, does domain-specific fine-tuning improve in-domain translation quality over a general-domain model and over a zero-shot baseline? Third, is continued fine-tuning from a general-domain checkpoint preferable to direct domain-specific supervised fine-tuning (SFT)?

We study two specialized domains that represent contrasting quality criteria. In the **literary** domain, translation quality depends on fluency, creativity, stylistic naturalness, and preservation of register and narrative voice. In the **clinical** domain, quality is primarily determined by terminological accuracy, factual completeness, and clinical adequacy. These differences motivate distinct evaluation emphases and are expected to affect domain learnability under limited synthetic supervision.

Our experimental pipeline constructs domain-specific synthetic corpora from available Catalan and Basque monolingual and bilingual resources. Using back-translation, we preserve the original monolingual texts as training targets and generate synthetic source-side data for fine-tuning. We adapt Latxa, a Basque-centric instruction-tuned large language model, with parameter-efficient LoRA adapters. We evaluate six trained model variants

¹<https://huggingface.co/HiTZ/Latxa-Qwen3-VL-8B-Instruct>

and a zero-shot baseline across four automatic metrics (chrF++, BLEU, TER, and COMET) and include a controlled token-matched comparison to assess domain learnability independently of data volume.

2 Related work

Domain adaptation for MT has been extensively studied in resource-rich settings, commonly through continued training on in-domain parallel data (Freitag and Al-Onaizan, 2016), mixture-of-domains training (Ling et al., 2025), and through fine-tuning of pre-trained language models (Alves et al., 2023). Back-translation has been widely explored as a method for leveraging monolingual data to generate synthetic bilingual corpora and improve MT performance (Luu et al., 2026; Bojar and Tamchyna, 2011; Edunov et al., 2018), including in low-resource language settings (Xu et al., 2019).

The Projecte AINA initiative (Villegas, 2023) at the Barcelona Supercomputing Center has produced dedicated MT models for Catalan, including a Basque–Catalan system² trained on data generated by translating the Spanish side of a CA–ES corpus through a Latxa ES–EU pivot system. That model is the closest prior system to the present work, differing in that it uses an encoder–decoder architecture trained from scratch rather than fine-tuning an instruction-tuned LLM.

As regards to Basque-centric language modelling, Latxa (Etxaniz et al., 2024) represents the state of the art for open Basque LLMs and is the foundation of the Latxa-Qwen3 and Latxa-Llama model series used in this work.

Supervised fine-tuning (SFT) is one of the most common approaches to domain adaptation in LLMs, enabling models to continue training on in-domain examples and specialize in a target domain (Alves et al., 2023; Eschbach-Dymanus et al., 2024). Nevertheless, applying full SFT to LLMs is costly, as it involves updating billions of model parameters. Parameter-efficient fine-tuning methods, in particular LoRA (Hu et al., 2022), have made it feasible to adapt large models with limited compute where full fine-tuning is expensive.

3 Data and synthetic corpus construction

We use four source corpora spanning three domains. Corpora exceeding 100,000 samples are randomly

²<https://huggingface.co/projecte-aina/aina-translator-eu-ca>

subsampled to at most 100,000 in order to control training size and reduce domain imbalance.

General domain. The AINA CA–EU Parallel Corpus³ provides general-domain sentence pairs. We sample 50,000 sentence pairs. Each pair is used bidirectionally: one EU→CA and one CA→EU instruction sample are generated per pair, giving up to 100,000 instruction instances before filtering and the train/ validation/test split.

Literary A (CTILC). The original Catalan texts come from the CTILC Catalan literary corpus (Institut d’Estudis Catalans, 2015). We rely on a previously constructed synthetic Catalan–Spanish corpus (Guerrero Castelló, 2026), where the CTILC Catalan paragraphs were translated into Spanish with Gemma-3-27B-IT, resulting in approximately 30,000 aligned Catalan–Spanish paragraph pairs. For our CA–EU corpus construction, we use Spanish as a pivot language and translate the synthetic Spanish side into Basque with Latxa-Llama-3.1-8B-Instruct. This produces synthetic Basque sources paired with original Catalan targets. The resulting pairs are then aligned using multilingual-e5-large embeddings.

Literary B (EhuHac). The EhuHac Spanish–Basque literary corpus⁴ provides sentence-aligned EU–ES pairs. Spanish is translated into Catalan, producing synthetic CA–EU literary pairs. Sentence pairs in which either side is shorter than five characters are discarded. From the remaining valid pairs, we randomly sample 100,000 examples.

Clinical (E3C). The E3C Basque clinical corpus⁵ provides approximately 1,300 document records. Basque clinical paragraphs are translated into Catalan using Latxa-Llama-3.1-8B-Instruct. Because clinical samples are comparatively longer, documents are split into chunks of at most 2,000 characters before translation.

3.1 Synthetic data construction via back-translation

We generate synthetic parallel data from target-side monolingual or pivoting Spanish texts using Latxa-Llama-3.1-8B-Instruct⁶. This model was used for

³https://huggingface.co/datasets/projecte-aina/CA-EU_Parallel_Corpus

⁴<https://opus.nlpl.eu/datasets/EhuHac?pair=eu&es>

⁵<https://github.com/hltfbk/E3C-Corpus>

⁶<https://huggingface.co/HiTZ/Latxa-Llama-3.1-8B-Instruct>

Domain	Direction	Train		Validation		Test	
		<i>total</i>	<i>per dir.</i>	<i>total</i>	<i>per dir.</i>	<i>total</i>	<i>per dir.</i>
General	EU→CA	87,229 (90%)	43,568 (49.9%)	4,846 (5%)	2,458 (50.7%)	4,847 (5%)	2,435 (50.2%)
	CA→EU		43,661 (50.1%)		2,388 (49.3%)		2,412 (49.8%)
Literary	EU→CA	102,344 (90%)	26,900 (26.3%)	5,685 (5%)	1,468 (25.8%)	5,687 (5%)	1,426 (25.1%)
	CA→EU		75,444 (73.7%)		4,217 (74.2%)		4,261 (74.9%)
Clinical	CA→EU	1,174 (90%)	1,174	65 (5%)	65	66 (5%)	66

Table 1: Dataset splits by domain and direction. Percentages in the *total* columns are computed over the full domain size, while percentages in the *per dir.* columns are computed within each split.

corpus construction because it was the available Latxa-family instruction-tuned model at the time the synthetic translation pipeline was run. The fine-tuning experiments themselves are conducted on Latxa-Qwen3-VL-8B-Instruct, which was selected later due to its reported adaptation to Basque, Catalan, and Galician. In all cases, we use the back-translation technique: the machine-translated text is always used as the *source* side during fine-tuning, while the original human-authored text is preserved as the *target* side. In this way, the model is trained to produce clean, human-quality output from potentially noisy synthetic input, rather than learning to imitate machine-translated targets. Table 2 summarizes the construction direction for each corpus.

Corpus	Source (synthetic)	Target (original)	Dir.
CTILC literary	EU (translated)	CA (original)	EU→CA
EhuHac literary	CA (translated)	EU (original)	CA→EU
E3C clinical	CA (translated)	EU (original)	CA→EU

Table 2: Round-trip supervision: synthetic source, original target.

3.2 Dataset splits

As shown in Table 1, all domains use a 90/5/5% partition split. The difference of sizes in the literary training set (CA→EU: 75,444 vs. EU→CA: 26,900) reflects the imbalance of the two source corpora: EhuHac provides substantially more sentence pairs for the CA→EU direction than CTILC provides for EU→CA. The clinical corpus covers only CA→EU because no Catalan clinical originals are available; all Catalan text is synthetic. Moreover, the large average paragraph length in the clinical domain ($\sim 1,078$ tokens) contrasts sharply with the literary domain (~ 110 tokens) and explains the chunking requirement in the translation pipeline.

4 Methodology

4.1 Base model

All fine-tuning experiments use Latxa-Qwen3-VL-8B-Instruct as the base model.

4.2 Instruction format

We formulate fine-tuning as an instruction-following translation task, detailed in Table 3. Consistently with our back-translation setup, the direction is determined by the original side: Catalan originals are used as CA targets for EU→CA fine-tuning, while Basque originals are used as EU targets for CA→EU fine-tuning, thus preserving maximum quality of the translations in the target.

Setting	Instruction prefix
EU→CA general	<i>Itzuli testu hau euskaratik katalanera:</i>
CA→EU general	<i>Tradueix aquest text del català al basc:</i>
EU→CA literary	<i>Itzuli testu literario hau euskaratik katalanera:</i>
CA→EU literary	<i>Tradueix aquest text literari del català al basc:</i>
CA→EU clinical	<i>Tradueix aquest text clínic del català al basc:</i>

Table 3: Instruction prefixes used for each translation direction and domain.

4.3 Fine-tuning strategy

We use parameter-efficient fine-tuning with LoRA adapters. Following a fixed configuration across all experiments, we set the LoRA rank to $r = 16$, the scaling factor to $\alpha = 32$, and the dropout rate to $p = 0.05$. The adapters are inserted into the attention projections (q_proj, k_proj, v_proj, o_proj) and the MLP projections (gate_proj, up_proj, down_proj). Additionally, for memory-efficient training, the base model is loaded in 4-bit NF4 quantization with bitsandbytes.

4.4 Training hyperparameters

Table 4 lists the shared hyperparameters for training. The maximum sequence length is set to

768 tokens, which covers approximately the 95th percentile of the training data while limiting memory consumption during fine-tuning. Checkpoint selection is based on best validation BLEU. Early stopping with patience 3 is applied where the validation set is large enough to be meaningful.

Hyperparameter	Value
Base model	Latxa-Qwen3-VL-8B-Instruct
LoRA rank / α	16 / 32
LoRA dropout	0.05
Quantization	4-bit NF4
Max sequence length	768 tokens
Epochs	3
Batch size	4
Gradient accumulation	8 steps
Effective batch size	32
Learning rate	5×10^{-5}
LR schedule	cosine
Warmup ratio	0.05
Checkpoint selection	best validation BLEU
Seed	42

Table 4: Shared training hyperparameters for all model variants.

5 Experimental setup

5.1 Model variants

We compare six trained model variants against a zero-shot baseline, including one token-matched variant for the learnability analysis. Table 5 provides an overview.

Variant	Description
Baseline	Zero-shot Latxa-Qwen3-VL-8B-Instruct
generalv1	Direct SFT on AINA general CA-EU data
literaryv1	Direct literary SFT from base model
literaryv2	Literary SFT continued from generalv1
clinicalv1	Direct clinical SFT from base model
clinicalv2	Clinical SFT continued from generalv1
litv1-tm	CA→EU literary SFT, token-matched to clinical

Table 5: Model variants and their training lineage. litv1-tm denotes the token-matched literary variant.

For the v2 variants, we perform continued domain adaptation from the general-domain checkpoint. The LoRA adapters trained in generalv1 are first merged into the base model weights, and a new set of LoRA adapters is then trained on the corresponding domain-specific corpus. This setup tests whether starting from a general CA-EU translation model improves specialized adaptation. Since the general-domain corpus consists

of existing parallel data rather than synthetic data generated in our pipeline, we expect it to provide a cleaner and higher-quality initialization than the domain-specific synthetic corpora.

5.2 Token-matched experiment

To compare literary and clinical domain learnability independently of data volume, we train litv1-tm on a CA→EU literary subset matched to the clinical training budget in source tokens, approximately 360,000 source tokens. We restrict the literary subset to the same translation direction as the clinical corpus, CA→EU, so that both models are trained and evaluated under comparable directional conditions. We conduct token matching instead of sample matching owing to the large difference in average length per sample between the datasets of the two domains. In the literary-domain dataset, each sample corresponds to a sentence, whereas in the clinical-domain dataset, each sample corresponds to a paragraph. As a result, clinical samples are approximately ten times longer on average, as shown in Table 6.

Domain	Average length	Δ
Literary	110 tokens	$1.0\times$
Clinical	1,078 tokens	$\sim 9.8\times$

Table 6: Average sample length by domain.

Both litv1-tm and clinicalv1 are evaluated on the original test sets of their respective domains (detailed in Table 1).

5.3 Evaluation

For evaluation, we generate translations with each fine-tuned model and compare them against the reference translations. We report chrF++, BLEU, TER and COMET (Popović, 2015; Papineni et al., 2002; Snover et al., 2006; Rei et al., 2022). TER is expressed as a percentage throughout.

- **chrF++:** used to capture character-level overlap, which is particularly relevant for agglutinative languages such as Basque, as it is more sensitive to subword and morphological variation.
- **BLEU:** included as a standard MT benchmark metric, allowing comparison with previous work despite its limitations for morphologically complex languages and low-resource settings.

- **TER**: Translation Edit Rate measures the number of edits needed to transform the system output into the reference translation, hence lower values indicate better output.
- **COMET**: included to complement surface-overlap metrics with a learned quality signal that is more sensitive to adequacy and semantic similarity.

Bidirectional evaluation (EU→CA and CA→EU) is performed for the general and literary domains; clinical evaluation covers only CA→EU.

6 Results

6.1 General domain

As shown in Table 7, the zero-shot baseline produces near-zero BLEU and very high TER in both directions, confirming that the base model’s multilingual capacity alone is insufficient for this low-resource pair despite its Basque–Catalan coverage. In contrast, the generalv1 evidences that fine-tuning on only 87,000 general-domain sentence pairs yields very large gains: chrF++ rises by 35 points for EU→CA and by 24 points for CA→EU, BLEU improves by roughly 18 for EU→CA and by 9 points for CA→EU respectively, and COMET increases more than 30 points in both directions. This demonstrates that the base model has the necessary linguistic competence but requires task-specific supervision to produce usable translations for this pair.

Model	Dir.	chrF++ ↑	BLEU ↑	TER ↓	COMET ↑
Baseline	EU→CA	10.93	0.60	300.3	45.17
Baseline	CA→EU	17.10	1.29	296.3	48.54
generalv1	EU→CA	45.03	18.51	75.53	80.61
generalv1	CA→EU	41.63	9.92	88.02	80.75

Table 7: General domain results. Bold: best per direction.

6.2 Literary domain

Model	Dir.	chrF++ ↑	BLEU ↑	TER ↓	COMET ↑
Baseline	EU→CA	11.96	0.82	151.3	41.55
Baseline	CA→EU	16.59	0.63	266.6	44.82
generalv1	EU→CA	27.97	4.96	89.44	66.68
generalv1	CA→EU	29.58	2.82	101.8	67.76
literaryv1	EU→CA	36.13	8.96	85.08	69.61
literaryv1	CA→EU	26.90	2.44	99.60	65.29
literaryv2	EU→CA	34.51	7.44	87.66	68.72
literaryv2	CA→EU	25.87	2.31	100.7	64.34

Table 8: Literary domain results. Bold: best per direction.

As shown in Table 8, the generalv1 model transfers to the literary domain substantially better than the zero-shot baseline, confirming that general-domain CA→EU alignment is a useful starting point.

If one compares the general-domain model versus the direct literary SFT model (literaryv1), there are mixed results, but literaryv1 is better overall. The literaryv1 model robustly outperforms generalv1 in EU→CA (chrF++ rises from 27.97 to 36.13), whereas the generalv1 model is slightly better in CA→EU.

Regarding domain-specific variants, literaryv1 slightly but consistently outperforms the continued adaptation variant literaryv2 across all reported metrics and both directions. This at first sight looks like a counterintuitive finding: continued fine-tuning from a strong general-domain starting point does not surpass direct literary SFT. One possible explanation is that the general checkpoint encodes translation patterns from distinct domains that introduce noise during literary adaptation, whereas direct SFT from the base model allows the LoRA adapters to specialize more cleanly to literary style. An alternative explanation is that the literary synthetic data already contains sufficient signal for direct adaptation, making the general checkpoint an unnecessary intermediate step.

BLEU scores in the CA→EU direction are notably low across all literary variants. This is partly expected: Basque is a highly agglutinative language, and BLEU considers the full word incorrect even if most of its morphemes are correct. chrF++ and COMET are more informative in this setting.

6.3 Clinical domain

Model	chrF++ ↑	BLEU ↑	TER ↓	COMET ↑
Baseline	11.03	0.64	104.3	50.56
generalv1	10.60	0.38	98.10	56.56
clinicalv1	40.20	19.43	101.09	76.25
clinicalv2	38.73	18.50	104.64	75.02

Table 9: Clinical domain results (CA→EU only). Bold: best value per metric.

As can be observed in Table 9, the clinical domain exhibits the most drastic gains from the baseline: clinicalv1 raises chrF++ from 11.03 to 40.20 and BLEU from 0.64 to 19.43.

The general-domain model applied cross-domain to the clinical test set (generalv1) yields chrF++ below the zero-shot baseline (11.03 vs.

10.60) and lower BLEU, suggesting that general-domain fine-tuning may orient the model towards general-domain fluency patterns that are actually less appropriate for clinical terminology.

As with the literary domain, direct SFT (clinicalv1) slightly but consistently outperforms continued fine-tuning from the general checkpoint (clinicalv2) across all metrics.

In an overall comparison of all models, clinicalv1 is the strongest system by three out of four metrics (chrF++, BLEU, and COMET), whereas generalv1 achieves the best TER score.

The clinical test set contains only 66 examples, so numerical differences should be interpreted with caution. Nevertheless, the consistency of the ranking across three metrics (chrF++, BLEU, COMET) lends some confidence to the conclusion.

6.4 Token-matched comparison

Model	chrF++ ↑	BLEU ↑	TER ↓	COMET ↑
clinicalv1	40.20	19.43	101.09	76.25
litv1-tm	24.73	2.28	97.03	65.41

Table 10: Token-matched CA→EU results. Both models are trained on approximately 360,000 source tokens and evaluated on their respective domain test sets.

In Table 10, one can observe that, when trained on equal token budgets, clinical SFT achieves substantially higher chrF++ (40.20 vs. 24.73), BLEU (19.43 vs. 2.28), and COMET (76.25 vs. 65.41) than literary SFT, while TER is slightly better for the literary model (litv1-tm). This suggests that clinical texts are intrinsically more learnable under the same experimental conditions. Clinical language tends to be formulaic, with recurring terminology, standardized phrase structures, and unambiguous semantics. Literary translation, in contrast, requires fluency, stylistic variation, figurative language, and narrative coherence (properties that are harder to capture with a synthetic training set and that automatic metrics may inadequately quantify). The gap in chrF++ is particularly large, consistent with the greater morphological predictability of clinical neologisms, along with the hardly predictable lexical diversity of literary texts.

7 Discussion

Pivot translation and back-translation. This work demonstrates that, in the complete absence

of direct CA→EU in-domain parallel corpora, pivot-based translation (via Spanish) and back-translation from monolingual data can produce training signal sufficient for substantial improvement. The approach is practical: it requires only a multilingual model capable for data construction and it does not depend on human translators for domain-specific pairs.

However, synthetic data introduces risks. Machine-translated source sides may contain *translationese* artifacts, pivot language biases, or domain-specific mistranslations. The back-translation strategy (training the model to produce original human-authored targets from synthetic source inputs) mitigates target-side noise but cannot eliminate source-side errors.

Direct SFT vs. continued adaptation. In both literary and clinical domains, direct SFT from the base model outperforms continued fine-tuning from the general-domain checkpoint. This is contrary to our initial expectation: because the general-domain corpus consists of existing parallel data rather than synthetic data generated in our pipeline, we expected it to provide a cleaner starting point for specialized adaptation.

A possible explanation is that the general-domain checkpoint encodes translation patterns that are not fully aligned with the stylistic or terminological requirements of the specialized domains. Direct SFT may therefore allow the LoRA adapters to specialize more cleanly from the base model, while the domain-specific synthetic corpora already provide enough data for adaptation without an intermediate step that may introduce additional noise.

Metric considerations. The four metrics provide complementary views of translation quality, but their interpretation requires caution in this setting. chrF++ is particularly useful for Basque because it captures subword overlap, while BLEU may underrepresent partially correct morphological forms. TER is informative about editing effort and re-ordering, which is relevant given the typologically different word orders of Catalan (SVO) and Basque (SOV). COMET provides a learned adequacy signal and correlates better with human judgement in many settings, but it is trained primarily on high-resource language pairs and its performance on CA→EU may not be optimal. All four metrics should be treated as complementary rather than

individually. This limitation is especially important in the absence of human evaluation: literary style and clinical adequacy cannot be fully captured by these metrics trained primarily on other language pairs.

Cross-domain transfer. The general-domain model used cross-domain shows mixed behavior: it improves over the zero-shot baseline in the literary domain (as expected from its CA–EU alignment) but decreases chrF++ and BLEU in the clinical domain. This suggests that general-domain patterns can hinder clinical adaptation at the lexical level as terminology is highly specialized.

8 Limitations and future work

Several limitations should be acknowledged. First, the clinical setting is limited by the small size of the test set ($N = 66$), so clinical results should be interpreted with caution. In addition, clinical training covers only the CA→EU direction: the reverse direction is unavailable due to the absence of Catalan clinical original datasets. Second, the study relies heavily on synthetic supervision. Although our back-translation setup preserves human-authored texts as targets, the synthetic source side may still contain translationese artifacts, pivot biases, or mistranslations. Third, all experiments are conducted with a single model family, Latxa-Qwen3, which limits the extent to which the findings can be generalized to other architectures. Finally, evaluation is entirely automatic: human assessment of translation quality could be particularly useful for the literary style.

Future work should address these limitations and integrate other attempts for improvement. First, evaluating the approach on human-translated parallel data would help determine how much of the observed performance is affected by synthetic-data noise. Second, English-language instruction prefixes (instead of the implemented source-language instructions) should be explored, as researchers leveraging the Latxa-Qwen3-VL-8B-Instruct model are experiencing better results with English prompting (Josu Barrutia, personal communication, 2026). Additionally, ablation studies should test different pivot-language choices, training-data sizes, and LoRA configurations in order to identify which factors contribute most to adaptation quality.

9 Conclusion

We have presented a domain adaptation study for Catalan–Basque machine translation covering general, literary, and clinical domains, using synthetic bilingual data derived from monolingual and pivot-language resources. Fine-tuning Latxa-Qwen3-VL-8B-Instruct with LoRA adapters yields very large improvements over the zero-shot baseline in all domains. Domain-specific fine-tuning consistently outperforms the general-domain model applied cross-domain, and direct SFT outperforms continued fine-tuning from the general checkpoint in both literary and clinical settings. The token-matched experiment reveals that clinical texts are more learnable than literary texts, probably because clinical texts are stylistically more similar to the standard-register texts on which LLMs are mostly pretrained. Findings confirm that pivot-based and back-translated synthetic data can compensate effectively for the complete absence of direct CA–EU in-domain parallel corpora, while supporting the effectiveness of back-translation and domain-specific fine-tuning for low-resource specialized MT.

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A Resources

The code used in this work is publicly available at:

- <https://github.com/pguerrero-igutierrez/MT-domain-adaptation>

The fine-tuned model variants are released through the following Hugging Face collection:

- <https://huggingface.co/collections/pguerrero-igutierrez/mt-domain-adaptation-ca-eu>